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Exp 1 :- Caesar Cipher

Aim :- To write a c program to implement the caesar cipher algorithm by shifting each other letters of the plaintext by k positions.

Algorithm :-

1. Start
2. Read the plaintext and the key (k)
3. For each charecter :
   - If it is an UC letter, shift it by k positions
   - If it is a lowercase letter, shift it by k position.
   - Leave non-alphabet charecters unchanged.
4. Display the encrypted text
5. Stop

Output :- Plaintext : Hello

key : 3

Encryted text : KHOOR

Result :- thus the program was executed successfulley.

Exp 2:- Monoalphabetic Substitution Cipher

Aim:- To write a C program to implement the mono-alphabetic substitution cipher using a substitution alphabet.

Algorithm:-

1. Start
2. Read the plaintext
3. Define a substitution alphabet
4. Replace each plaintext letter with its corresponding cipher letter.
5. Display the encrypted text
6. Stop.

Output:- Plaintext :- HELLO

Encrypted text : ITSSG

Result:- Thus, the program was executed successfully.

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Exp 3 :- Playfair Cipher.

Aim:- To write a c program and implement the playfair cipher using a 5x5 matrix generated from a keyword

Algorithm:-

1. Start
2. Read the keyword and plaintext
3. Construct the 5x5 matrix, using the keyword
4. Divide the plain text into pairs of letters.
5. Encrypt each pair according to playfair rules:
   - Same row → replace with next letter in the row.
   - Same column → replace with the next letter in column
   - Rectangle → replace with the letters at the opposite corners.
6. Display the encrypted text
7. Stop.

Output:-

Key : MONARCHY

Plaintext : HELLO

Encrypted text : CFSUPM

Result:- Thus, the program was executed Successfully.

Exp-4 :- Polyalphabetic Substitution Cipher

Aim :- To write a C program to Implement the Poly-alphabetic Substitution cipher using a keyword

Algorithm :-

1. Start
2. Read the plaintext and keyword
3. Repeat the keyword until it matches the plaintext length
4. Encrypt each letter using the vigenette formulae.
5. Display the encrypted text
6. Stop

Output :-

Plaintext : HELLO

KEyword : KEY

Encrypted text : RIJVS.

Result :- Thus the program was executed Successfully.

Exp 5: Affine Caesar Cipher

Aim:- To write a C program to Implement the Affine cipher using the formulae.

$$C = (aP + b) \bmod 26$$

Algorithm:-
1. Start
2. Read the plaintext
3. Read values of a and b
4. Verify that gcd (a,26) = 1
5. Encrypt every charecter using the Affine formulae
6. Find the multiplicative inverse of a
7. Decrypt the ciphertext
8. Display the results
9. Stop

Output:-

Plain Text : HELLO

a=5

b=8

Encryted Text : RCLLA

Decrypted Text : HELLO

Result:- Thus, the Affine Cipher was successfully implemented and the original message was recovered

Exp 6 :- Breaking and an Affine cipher

Aim:- To write a C program to break an Affine ciphe using frequency analysis

Algorithm:-
1. Start
2. Read the ciphertext
3. Find the frequency of each letter
4. Assume the most frequent letters correspond to 'E' and 'T'.
5. Solve for values of a and b
6. Find the multiplicatir inverse of a
7. Decrypt the ciphertext.
8. Display the plaintext
9. Stop

Output:- Most Frequent Letter : B

Second Frequent letter : U

calculated keys

a = 7

b = 21

Decrypted message : HELLO WORLD

Result:- Thus, the Affine Cipher was successfully broken using frequency analysis

Exp 7 : Decrypt a Simple Substitution Cipher

Aim:- To write a C program to decrypt a simple substitution cipher using frequency analysis.

Algorithm :-

1. Start
2. Read the Ciphertext
3. Count the occurence of each symbol
4. Identify the most frequent symbols
5. Guess common English letters (E, T, H)
6. Replace symbols gradually
7. Continue until the plaintext is obtained
8. Display the decrypted message
9. Stop

Output :-

Cipher Text:

53 # # + 305)) 6* ....

Decrypted Message:

A Good Glass in the Bishop's Hostel.

Result :- Thus, the ciphertext was successfully decrypted using frequency analysis.

Exp 8 :- Keyword Monoalphatic Cipher

Aim:- To write a C program to Implement a Keyword monoalphabetic substitution cipher using the keyword CIPHER.

Algorithm:-
1. Start
2. Read the plaintext
3. Read the keyword
4. Construct the cipher alphabet using the keyword
5. Append the remaining unused letters
6. Replace each plaintext letter with its corresponding cipher text
7. Display the ciphertext
8. Decrypt to recover the original Plaintext
9. Stop.

Output:-

Keyword : CIPHER

Plaintext : COMPUTER

Encrypted : PMKOUTEQ

Decryted : COMPUTER

Result:- Thus, the keyword monoalphabetic cipher was successfully implemented using the keyword and the plaintext was encrypted & decryped correctly

Exp 9:- Playfair cipher Decrytion

Aim:- To write a C program to decrypt the given Playfair cipher message using the appropriate Playfair key matrix.

Algorithm:-
1. Start
2. Read the playfair ciphertext
3. Construct the 5x5 playfair matrix using the given keyword.
4. Divide ciphertext into pairs of letters.
5. Combine all decrypted pairs
6. Display the plaintext.
7. Stop

Output:-

Cipher text: KXJEY UREBE ZWEHE WRYTU...

Decrypted text: PT BOAT ONE NINE LOST IN ACTION.

Result:- Thus the playfair ciphertext was successfull decrypted using the playfair algorithm.

EXP 10: Playfair cipher Encryption.

Aim:- To write a C program to encrypt the message " Must see you over cadogan west. Coming at once" using the given playfair matrix.

Algorithm:-
1. Start
2. Use the given playfair matrix
3. Remove spaces and, Punctuation.
4. Convert all letters to uppercase
5. Replace J with I
6. Divide the plaintext into digraphs
7. Insert x between repeated letters if requid.
8. Apply playfair encryption rules.
9. Display the ciphertext
10. STOP

Output:-

Plain text: MUST SEE YOU OVER CAROGAN WEST COMING AT ONCE

Encrypted Text:-

QPNQXQY MV...

Result:- Thus, the plaintext was successfully encrypted using the given playfair matrix.

Exp 11 :- Number of possible Playfair Keys

Aim:- To write a C program to determine the number of possible keys available in the playfair cipher.

Algorithm:-

1. Start
2. Consider the 25 - letter playfair matrix
3. Calculate the total permutations as 25!
4. Express the result as an approxima power of 2
5. Display the total possible keys.
6. STOP

Output:-

Total possible keys = 25!

Approximate value = 2^84.

Part (a) :

Effective unique keys = 25! /2.

Result:- Thus, the apporoximate numbeo of possibl Playfair key was successfully determined.

Exp 12. Hill Cipher Encryption & Decryption

Aim: To write a C program to encrypt the plaintext "meet me at the usual place at ten rather than eight o clock" using the Hill cipher with the given key matrix.

Key matrix: 9 4
5 7

Algorithm:-

1. Start
2. Read the plaintext
3. Remove spaces
4. Convert letters into Numericals
5. Divide the plaintext into pairs
6. Multiply each pair with the key matrix
7. Apply modulo 26 & convert back to letters.
8. Display and STOP

Output:-

Plain text MEETMEATTHEUSUALPLACE

Encrypted text : UKIX UKYDROMEI

Decrypted text : MEETMEATTHE USUAL PLACE

Result:- Thus, the Hill cipher was successfully encrypted & Decrypted using inverse key matrix.

Exp 13: Known Plaintext Attack on Hill Cipher

Aim:- To write a C program to demonstrate a known-plaintext attack on the Hill cipher

Algorithm :- 1. Start

2. Read known plaintext and corresponding ciphertext
3. Convert both into matrix form.
4. Compute the inverse of the plaintext matrix
5. Multiply the ciphertext matrix with the inverse plaintext matrix
6. Recover the key matrix
7. Display the recovered key.
8. STOP

Output:- Known plain text:

HELP

Cipher text:

ZEBR

Recovered key matrix:

9 4
5 7

Result:- Thus, the Hill cipher key was successfully recovered using the known plaintext attack.

Exp 14: One - Time Pad Encryption & Decryption

Aim:- To write a 'C' program to encrypt the Plaintext "SEND MORE MONEY" using the one-Time Pad (OTP) algorithm.

Key stream:- 9 0 1 7 23 15 21 14 11 11 2 8 9

Algorithm:-
1. Start
2. Read the plaintext
3. Convert letters into numbers
4. Read the random key
5. Add the key value to each plaintext letter.
6. Apply modulo 26
7. Display ciphertext
8. STOP

Output:-

Plain Text: SEND MORE MONEY

cipher text: BNOH

Disired plain text: CASH NOT NEEDED

Generated key: 7 14 11 4 15

Result:- Thus, the required one-Time Pad key was successfully generated.

Exp 15 :- Letter Frequency Attack on Additive cipher

Aim :- To write a C program that performs a letter frequency attack on a additive (caesar) cipher without human Intervention

Algorithm :- 1. Start

2. Read the Ciphertext
3. Count the frequency of each letter
4. Assume the most frequent letter
5. Try all possible shift values (0-25)
6. Generate possible plaintexts
7. Rank the Plaintexts according to English Letter frequency.
8. Display the most likely plaintext
9. STOP.

Output :- CIPHER Text : KHOOR ZRUOG

Possible plain texts : shift 3 : HELLO WORLD

Shift 5 : FCJJM UMPJB

Best Match : HELLO WORLD

Result :- Thus, the additive cipher was successfu attacked using automatic letter frequency analysis, and the most probable PT was Ident